SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2_AT3 — CASE STUDY

Course Code:	DSA0606	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO2 – Evaluation (BL5)	**Assessment:**	Assessment Tool 3 – Case Study
Weightage:	20% of CIA	**Total Marks:**	20 (2 Questions × 10 Marks)
CO2 Statement:	*Analyze real-world data scenarios and evaluate/justify the most appropriate chart types, coordinate systems, color scales, and dashboard designs to represent them.(BL5)*		
Student Name:	P.Vishwa Nandan	**Reg. No.:**	192473015
Section / Batch:	2024	**Date:**	01-08-2026

Code of Conduct

I P.Vishwa Nandan *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: vishwanandanreddy

Instructions

- This test contains 2 case-study questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each case study has four sub-parts (a–d); answer all parts for full marks.
- Support your answers with brief calculations or justifications where relevant.
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – CASE STUDY

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT3 — Case Study

CO2 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO2-AT3 Case Study questions, in which students analyze a realistic multi-part scenario and select, justify, design, and critique appropriate visualizations across chart types, coordinate systems, color scales, and dashboard development.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough understanding of the full range of chart types, coordinate systems, and color scales relevant to the case, and correctly identifies which apply to each sub-part.	Good understanding of the relevant concepts, with only minor gaps in identifying appropriate chart types, scales, or color choices.	Basic understanding, but with some misconceptions about which visualization techniques suit the scenario's data.	Poor understanding of core concepts; proposes techniques fundamentally unsuited to the case study's data.
Explanation	**25%**	Provides detailed, well-reasoned justification for every design decision and critique, explicitly tied to specific details of the case.	Provides clear justification for most decisions and critiques, with adequate reasoning.	Justification is limited or generic; decisions are stated without sufficient reasoning tied to the case.	Little or no justification provided; answers are unsupported assertions.
Application	**20%**	Effectively applies visualization concepts to analyze and solve every sub-part of the case, including any calculations, with well-reasoned	Applies concepts correctly to most sub-parts, with only minor errors.	Limited application; some sub-parts are weakly addressed or only partially correct.	Fails to apply appropriate concepts; sub-parts are left unaddressed or answered incorrectly.

		conclusions.			
Organization	**15%**	The response is clearly structured, addressing every sub-part (a–d) in a logical sequence with coherent overall flow.	The response is organized and addresses each sub-part, with only minor issues in flow or clarity.	Basic structure; sub-parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and sub-parts are left unaddressed or answered out of order.
Accuracy	**10%**	All parts of the answer — chart choices, calculations, critiques, and dashboard design — are completely accurate and well-suited to the case.	Mostly accurate, with only minor omissions or a small error in one sub-part.	Partially correct; some elements are missing, mismatched to the case, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple sub-parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Name: P. Vishwa Nandan

Reg no.192473015

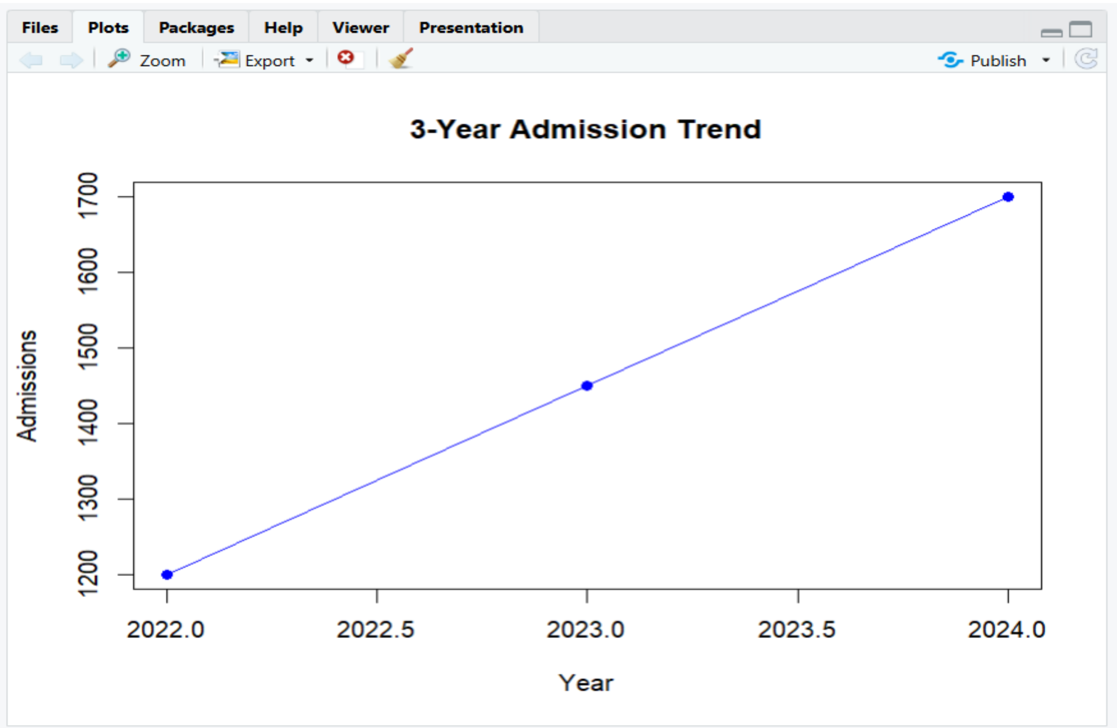
2(a) Which chart would best reveal the shape and outliers of the wait-time distribution across 20 departments? Justify your answer.

A Box Plot is the most suitable chart for revealing the shape and outliers of the wait-time distribution across 20 departments. A box plot summarizes the data using the minimum value, first quartile (Q1), median, third quartile (Q3), and maximum value, making it easy to understand how wait times are distributed. It also clearly identifies outliers, which are values that lie much higher or lower than the rest of the data. These unusual values are shown as individual points outside the whiskers of the box plot.

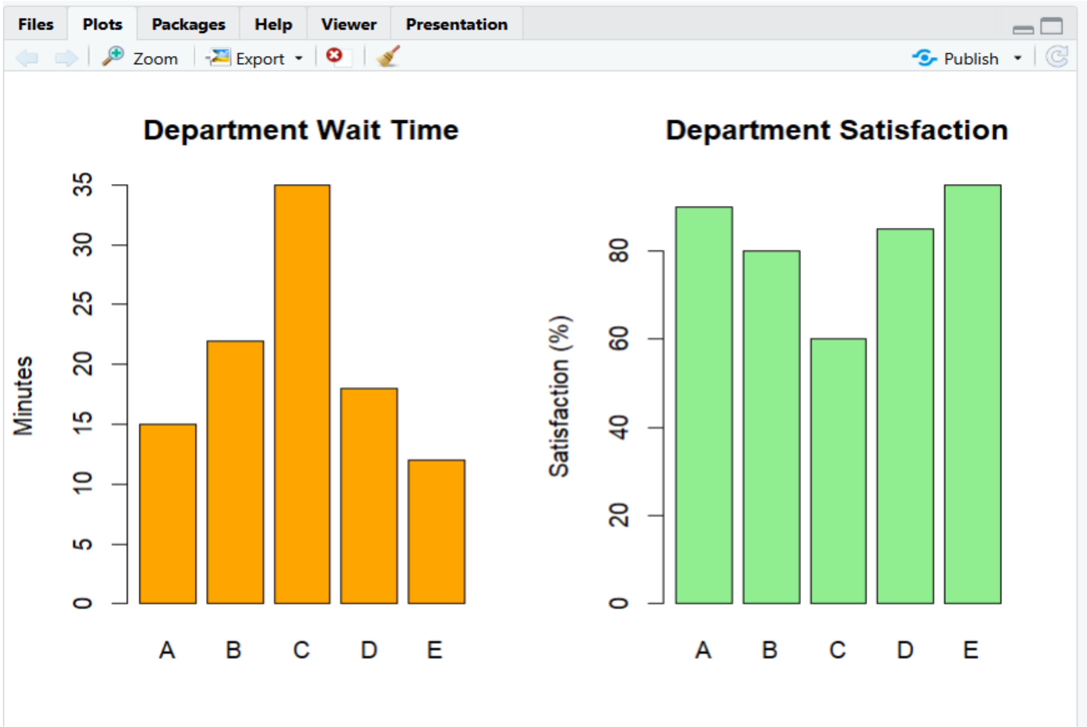
Using a box plot allows managers to compare the spread and consistency of wait times across departments. A department with a larger box or longer whiskers indicates greater variation in wait times, while a smaller box shows more consistent service. The position of the median line helps identify whether most departments have shorter or longer waiting times. The chart also gives an idea about the skewness of the data by showing whether the median is centered or closer to one end of the box.

Compared to a bar chart or pie chart, a box plot provides much more statistical information in a compact form. It is widely used in healthcare, education, and business analytics because it helps identify departments with unusually high waiting times that require immediate attention and supports better decision-making.

2b)



2c)



2(d) A colleague suggests a 20-slice pie chart to show each department's share of total wait time. Critique this choice and suggest a better alternative.

A 20-slice pie chart is not an appropriate choice for showing each department's share of the total wait time. Pie charts are most effective when displaying a small number of categories, usually fewer than six or seven. With 20 departments, the chart becomes overcrowded, making the slices too small to compare accurately. It is difficult for viewers to distinguish similar-sized slices or identify which departments contribute the most to the total wait time. Labels may overlap, and the chart becomes confusing rather than informative.

A better alternative is a horizontal bar chart or a sorted bar chart. A bar chart displays each department as a separate bar, making it much easier to compare wait times because people can accurately judge differences in bar lengths. Sorting the bars from highest to lowest wait time quickly highlights the departments with the longest waiting periods and those performing well. Labels are also easier to read, especially when department names are long.

Another advantage of a bar chart is that it can display exact values and can be enhanced with colors or annotations to highlight departments needing urgent improvement. Overall, a bar chart provides greater clarity, supports accurate comparisons, and communicates the information more effectively than a pie chart. Therefore, it is the preferred visualization for analyzing wait times across 20 departments.

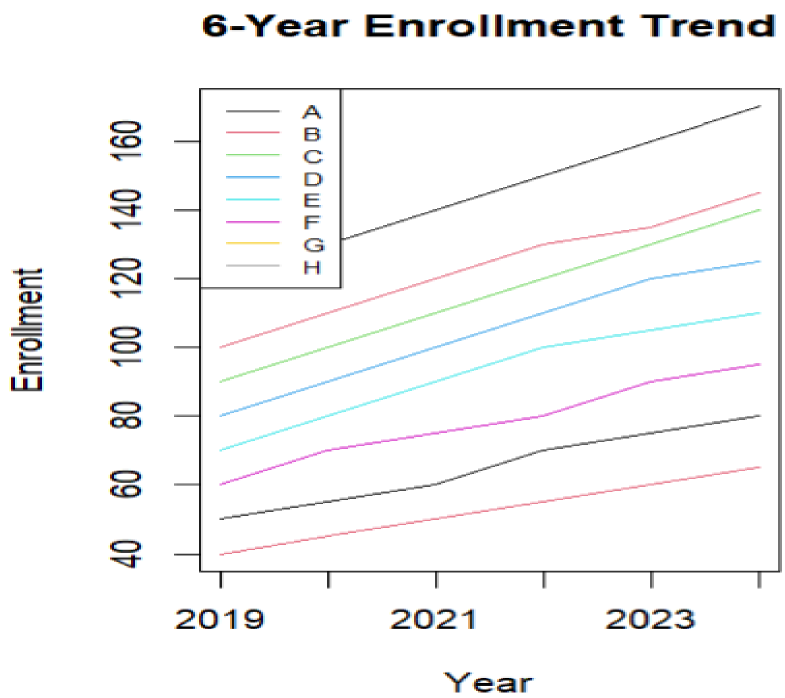
7(a) Which chart correctly preserves the order of the Low/Medium/High graduation categories? Justify your answer.

The most appropriate chart for preserving the order of the Low, Medium, and High graduation categories is an Ordered Bar Chart. These categories represent ordinal data, meaning they have a natural sequence from Low to Medium to High. An ordered bar chart displays the categories in this logical order, making it easy for viewers to understand the progression and compare the number of students in each category.

Unlike nominal categories, ordinal categories should never be arranged alphabetically or randomly because doing so would destroy their natural meaning. By placing the bars in the order Low → Medium → High, the chart clearly communicates the relationship between the categories. The height of each bar represents the number or percentage of students in that category, allowing easy comparison. Labels and values can also be added to improve readability.

Other charts, such as pie charts, show proportions but do not emphasize the natural ordering of categories. Similarly, line charts are mainly used for continuous or time-based data and are not suitable for ordinal categories. An ordered bar chart is simple, accurate, and easy to interpret. It enables viewers to quickly identify which graduation category has the highest or lowest number of students while preserving the meaningful order of the data, making it the best choice for this visualization.

7(b)



7(c) Which chart type best compares the 5 satisfaction aspects for a single department all at once? ***(~200 words)***

A Radar Chart (Spider Chart) is the best choice for comparing the five satisfaction aspects of a single department at the same time. A radar chart displays multiple variables on separate axes that extend outward from a central point. Each satisfaction aspect, such as teaching quality, facilities, staff support, placement assistance, and campus environment, is represented on one axis. The values are connected to form a polygon, providing an overall view of the department's performance.

The main advantage of a radar chart is that it allows viewers to evaluate all satisfaction aspects simultaneously rather than examining separate charts. Strong areas appear as points extending farther from the center, while weaker areas are closer to the center. This makes it easy to identify strengths and weaknesses at a glance. Decision-makers can quickly determine which aspects require improvement and which are performing well.

Alternative charts, such as bar charts, can also compare multiple aspects, but they require viewers to compare bar heights individually. A radar chart provides a more holistic visual summary of overall performance. It is especially useful when the number of variables is small, such as five satisfaction aspects. Therefore, a radar chart effectively presents multiple satisfaction measures in a single visualization, making it the most suitable chart for comparing the overall performance of one department.

7(d)